

Tutorial_Week2

Mingjie Wei

Setup

```
# You will need the correct file paths if you don't follow my naming conventions:
file_a <- "../texts/A07594__Circle_of_Commerce.txt"
file_b <- "../texts/B14801__Free_Trade.txt"

# Read the raw text files into R
text_a <- read_file(file_a)
text_b <- read_file(file_b)

# Combine into a tibble for tidytext workflows
texts <- tibble(
  doc_title = c("Text A", "Text B"),
  text = c(text_a, text_b)
)

texts
```

```
# A tibble: 2 x 2
  doc_title text
  <chr>      <chr>
1 Text A    THE CIRCLE OF COMMERCE. The Prooeme. HERODOTVS in his CLIO, reporte~
2 Text B    CAP. I. The Causes of the want of Money in England. IT hauing pleas~
```

Tokenization, stopwords, and counting words

1. Stopwords

```

# Start with tidytext's built-in stopwords list
data("stop_words")

# Add our own project-specific stopwords (you can, and will, expand this list later)
custom_stopwords <- tibble(
  word = c(
    "vnto", "haue", "doo", "hath", "bee", "ye", "thee"
  )
)

all_stopwords <- bind_rows(stop_words, custom_stopwords) %>%
  distinct(word)

all_stopwords %>% slice(1:10)

```

```

# A tibble: 10 x 1
  word
  <chr>
1 a
2 a's
3 able
4 about
5 above
6 according
7 accordingly
8 across
9 actually
10 after

```

2. Tokenization

```

word_counts <- texts %>%
  unnest_tokens(word, text) %>%
  mutate(word = str_to_lower(word)) %>%
  anti_join(all_stopwords, by = "word") %>%
  count(doc_title, word, sort = TRUE)

word_counts

```

```
# A tibble: 7,443 x 3
  doc_title word      n
  <chr>      <chr>   <int>
1 Text A    trade    232
2 Text A    exchange  186
3 Text B    trade    185
4 Text A    fo       184
5 Text A    malynes  158
6 Text A    merchants 126
7 Text A    mony     118
8 Text A    hee      115
9 Text A    kingdome 113
10 Text A   commodities 96
# i 7,433 more rows
```

Comparing word frequencies across texts

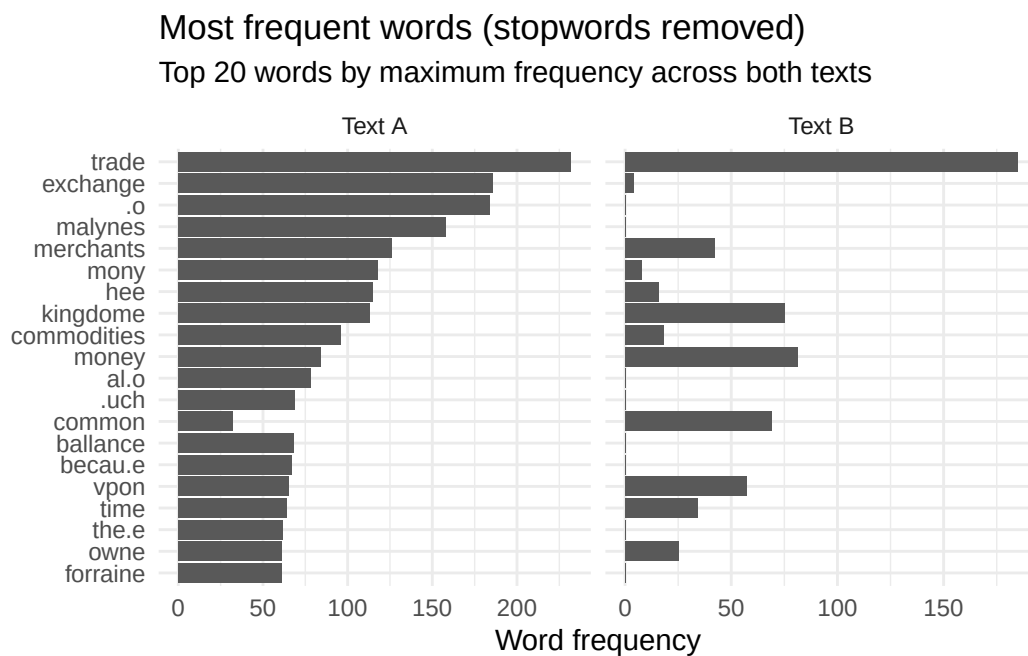
```
plot_n_words <- 20 # you can change this as needed

# Select the most frequent words overall
word_comparison_tbl <- word_counts %>%
  pivot_wider(
    names_from = doc_title,
    values_from = n,
    values_fill = 0
  ) %>%
  mutate(max_n = pmax(`Text A`, `Text B`)) %>%
  arrange(desc(max_n))

word_plot_data <- word_comparison_tbl %>%
  slice_head(n = plot_n_words) %>%
  pivot_longer(
    cols = c(`Text A`, `Text B`),
    names_to = "doc_title",
    values_to = "n"
  ) %>%
  mutate(word = fct_reorder(word, n, .fun = max))

ggplot(word_plot_data, aes(x = n, y = word)) + #black magic happens thanks to ggplot
  geom_col() +
  facet_wrap(~ doc_title, scales = "free_x") +
```

```
labs(
  title = "Most frequent words (stopwords removed)",
  subtitle = paste0(
    "Top ", plot_n_words,
    " words by maximum frequency across both texts"
  ),
  x = "Word frequency",
  y = NULL
) +
theme_minimal()
```



Bigrams: starting to think about context

- bigrams

```
bigrams <- texts %>%
  unnest_tokens(bigram, text, token = "ngrams", n = 2)

bigrams
```

```
# A tibble: 53,809 x 2
  doc_title bigram
```

```

      <chr>      <chr>
1 Text A      the circle
2 Text A      circle of
3 Text A      of commerce
4 Text A      commerce the
5 Text A      the prooeme
6 Text A      prooeme herodotvs
7 Text A      herodotvs in
8 Text A      in his
9 Text A      his clio
10 Text A     clio reportes
# i 53,799 more rows

```

- `separate()` takes the bigram column and splits each string at the space character.

```

bigrams_separated <- bigrams %>%
  separate(bigram, into = c("word1", "word2"), sep = " ")

bigrams_separated

```

```

# A tibble: 53,809 x 3
  doc_title word1      word2
  <chr>      <chr>      <chr>
1 Text A      the        circle
2 Text A      circle      of
3 Text A      of          commerce
4 Text A      commerce    the
5 Text A      the          prooeme
6 Text A      prooeme      herodotvs
7 Text A      herodotvs    in
8 Text A      in            his
9 Text A      his          clio
10 Text A     clio      reportes
# i 53,799 more rows

```

- remove `all_stopwords` (we remove bigrams where *either* word is a stopword, since phrases like “of the” or “and the” are rarely meaningful analytically.)

```

bigrams_filtered <- bigrams_separated %>%
  filter(
    !word1 %in% all_stopwords$word,
    !word2 %in% all_stopwords$word
  )

```

```
)
```

```
bigrams_filtered
```

```
# A tibble: 7,622 x 3
  doc_title word1      word2
  <chr>      <chr>    <chr>
1 Text A    prooeme  herodotvs
2 Text A    clio      reportes
3 Text A    croesvs   king
4 Text A    fonne     borne
5 Text A    borne     dumbe
6 Text A    king      himfelf
7 Text A    imminent danger
8 Text A    certaine perfian
9 Text A    perfian   ready
10 Text A    lay       violent
# i 7,612 more rows
```

- group by + count

```
bigram_counts <- bigrams_filtered %>%
  count(doc_title, word1, word2, sort = TRUE)
```

```
bigram_counts
```

```
# A tibble: 6,449 x 4
  doc_title word1      word2      n
  <chr>      <chr>    <chr>    <int>
1 Text B    common  wealth    48
2 Text A    merchants adventurers 35
3 Text B    latin   alphabet  34
4 Text A    low     countries 21
5 Text B    east    india     20
6 Text A    free    trade     19
7 Text A    latin   alphabet  17
8 Text A    native  commodities 17
9 Text A    forraine commodities 16
10 Text A    letters patents  14
# i 6,439 more rows
```

- Now that we removed the stopwords, we can use `unite` to put the bigrams back together

```
bigram_counts <- bigram_counts %>%
  unite(bigram, word1, word2, sep = " ")

bigram_counts
```

```
# A tibble: 6,449 x 3
  doc_title bigram          n
  <chr>      <chr>      <int>
1 Text B    common wealth    48
2 Text A    merchants adventurers 35
3 Text B    latin alphabet     34
4 Text A    low countries      21
5 Text B    east india         20
6 Text A    free trade         19
7 Text A    latin alphabet     17
8 Text A    native commodities  17
9 Text A    forraine commodities 16
10 Text A   letters patents    14
# i 6,439 more rows
```

Comparing bigrams

- normalize by the number of bigrams *within* each document to compare the two + transposition

```
bigram_relative <- bigram_counts %>%
  group_by(doc_title) %>%
  mutate(
    total_bigrams = sum(n),
    proportion = n / total_bigrams
  ) %>%
  ungroup()

bigram_wide <- bigram_relative %>%
  select(doc_title, bigram, proportion) %>%
  pivot_wider(
    names_from = doc_title,
    values_from = proportion,
    values_fill = 0
  )
```

```
bigram_wide
```

```
# A tibble: 6,379 x 3
  bigram          `Text B` `Text A`
  <chr>          <dbl>   <dbl>
1 common wealth  0.0187    0.00257
2 merchants adventurers 0      0.00692
3 latin alphabet 0.0133    0.00336
4 low countries  0.00156   0.00415
5 east india     0.00780    0
6 free trade     0      0.00376
7 native commodities 0.00273   0.00336
8 forraine commodities 0      0.00316
9 letters patents 0.000390  0.00277
10 thousand pounds 0      0.00277
# i 6,369 more rows
```

- We now want to know which bigram is most characteristic of “Text A” (*The Circle of Commerce*) relative to “Text B” (*Free Trade*).

```
bigram_diff <- bigram_wide %>%
  mutate(
    diff = `Text A` - `Text B`
  ) %>%
  arrange(desc(abs(diff)))

bigram_diff %>% slice(1:20)
```

```
# A tibble: 20 x 4
  bigram          `Text B` `Text A`   diff
  <chr>          <dbl>   <dbl>   <dbl>
1 common wealth  0.0187    0.00257 -0.0161
2 latin alphabet 0.0133    0.00336 -0.00989
3 east india     0.00780    0      -0.00780
4 merchants adventurers 0      0.00692  0.00692
5 merchants aduenturers 0.00390    0     -0.00390
6 free trade     0      0.00376  0.00376
7 forraine commodities 0      0.00316  0.00316
8 thousand pounds 0      0.00277  0.00277
9 low countries  0.00156   0.00415  0.00259
10 letters patents 0.000390  0.00277  0.00238
```


11 publique vtility	0.00234	0	-0.00234
12 cloth trade	0.00429	0.00198	-0.00231
13 33 fh	0	0.00218	0.00218
14 20 fhillings	0	0.00198	0.00198
15 fh 4	0	0.00198	0.00198
16 disorderly trade	0.00195	0	-0.00195
17 fishing vpon	0.00195	0	-0.00195
18 india stocke	0.00195	0	-0.00195
19 kings honour	0.00195	0	-0.00195
20 maiesties subiects	0.00195	0	-0.00195